

# Tutorial 1\_summary

## Tutorial 1 — Static Characteristics of Measurement System Elements

**Source:** Tutorial 1\_Solutions.pdf (17 pages)

**Instructor:** Qilong Zhong, MSc, GTIIT

**Corresponds to:** Lecture 1–2 (Chapters 1–2 of Bentley)

### 1. The General Measurement System

#### 1.1 Accuracy and Error

A perfectly accurate system is theoretical. Real system accuracy is quantified by measurement error  $E$ :

$$E = \text{Measured Value} - \text{True Value}$$

#### 1.2 Measurement System Structure

Process → [Sensing Element] → [Signal Conditioning] → [Signal Processing] → [Data Presentation] → Observer

### 2. Range and Span

| Concept      | Definition                    | Formula                                                              |
|--------------|-------------------------------|----------------------------------------------------------------------|
| <b>Range</b> | Allowable limits of operation | $I_{min} \rightarrow I_{max}, O_{min} \rightarrow O_{max}$           |
| <b>Span</b>  | Maximum variation             | Input Span = $I_{max} - I_{min}$ , Output Span = $O_{max} - O_{min}$ |

▲ Range describes operation limits, **not** accuracy or precision.

### 3. Ideal Straight Line

For a linear element:

$$O_{ideal} = KI + a$$

Where:

- $K = \frac{O_{max} - O_{min}}{I_{max} - I_{min}} = \textbf{slope}$  (sensitivity)
- $a = O_{min} - KI_{min} = \textbf{intercept}$  (zero offset)

✅ Units matter: If  $F$  is in Newtons and  $V$  is in Volts,  $K$  carries units of V/N;  $a$  has same units as the output.

## 4. Non-linearity

### Definition

If the actual output is  $O(I)$ , then:

$$N(I) = O(I) - (KI + a)$$

- $O(I)$  = actual output
- $KI + a$  = ideal straight-line output
- $N(I)$  = non-linearity deviation at input  $I$

### As Percentage of Full-Scale Deflection (FSD)

$$\% \text{ f.s.d.} = \frac{\hat{N}}{O_{max} - O_{min}} \times 100\%$$

where  $\hat{N} = \max |N(I)|$  is the **maximum** deviation.

▲ **Critical:** The denominator is the **output span**, NOT the input span.

## 5. Interfering Input ( $I_I$ )

**Definition:** An environmental variable that changes the **zero offset (intercept)** of the sensor output. It does NOT change the slope.

**Mathematical model:**

$$O = KI + a + K_I I_I$$

The **intercept** shifts from  $a$  to  $a + K_I I_I$ .

| Symbol | Meaning                                        |
|--------|------------------------------------------------|
| $K_I$  | Environmental sensitivity to interfering input |
| $I_I$  | Deviation from standard condition              |

**Example:** Supply voltage variation in a potentiometer — shifts the entire I/O curve up/down without changing sensitivity.

## 6. Modifying Input ( $I_M$ )

**Definition:** An environmental variable that changes the **sensitivity (slope)** of the sensor. It does NOT shift the intercept.

**Mathematical model:**

$$O = (K + K_M I_M)I + a$$

The **sensitivity** becomes  $K + K_M I_M$ .

| Symbol | Meaning                                        |
|--------|------------------------------------------------|
| $K_M$  | Change in sensitivity per unit modifying input |
| $I_M$  | Deviation from standard condition              |

**Example:** Temperature causes the spring stiffness in a pressure sensor to change → sensitivity changes, but zero-point stays fixed.

## 7. The Generalized Model (Summary)

Combining all effects:

$$O = KI + a + N(I) + K_M I_M I + K_I I_I$$

$$O = (K + K_M I_M)I + (a + K_I I_I) + N(I)$$

| Term        | Meaning                        |
|-------------|--------------------------------|
| $KI + a$    | Ideal straight line            |
| $N(I)$      | Non-linear deviation           |
| $K_M I_M I$ | Gain change due to environment |
| $K_I I_I$   | Zero shift due to environment  |

 **Engineering note:** To calculate  $K$ , we first set  $N(I) = 0$  (assume linearity over the calibration range).

## 8. ADC — Analog-to-Digital Converter

### Resolution

For an  $n$ -bit ADC:

$$\Delta V = \frac{V_{max} - V_{min}}{2^n}$$

$$\% \text{ f.s.d.} = \frac{1}{2^n} \times 100\%$$

| Bits | Levels   | Resolution (% f.s.d.) |
|------|----------|-----------------------|
| 8    | 256      | 0.39%                 |
| 10   | 1024     | 0.0976%               |
| 12   | 4096     | 0.0244%               |
| 16   | 65536    | 0.0015%               |
| 24   | 16777216 | 0.000006%             |

## 9. Hysteresis

**Definition:** Output depends on the **direction** of input change. For the same input value, output differs when input is increasing vs. decreasing.

$$H = V_{up} - V_{down}$$

$$\% \text{ f.s.d.} = \frac{H}{O_{max} - O_{min}} \times 100\%$$

## 10. Key Formulas Reference

| Formula                                                          | Application                |
|------------------------------------------------------------------|----------------------------|
| $O_{ideal} = KI + a$                                             | Linear element model       |
| $K = \frac{O_{max} - O_{min}}{I_{max} - I_{min}}$                | Sensitivity from endpoints |
| $a = O_{min} - KI_{min}$                                         | Intercept from endpoints   |
| $N(I) = O(I) - (KI + a)$                                         | Non-linearity deviation    |
| $\% \text{ ..} = \frac{\dot{N}}{O_{max} - O_{min}} \times 100\%$ | Non-linearity as %FSD      |
| $O = KI + a + K_I I_I$                                           | With interfering input     |
| $O = (K + K_M I_M)I + a$                                         | With modifying input       |
| $O = KI + a + N(I) + K_M I_M I + K_I I_I$                        | Generalized model          |

| Formula                                    | Application    |
|--------------------------------------------|----------------|
| $\Delta V = \frac{V_{max} - V_{min}}{2^n}$ | ADC resolution |
| $H = V_{up} - V_{down}$                    | Hysteresis     |

✅ **Key Takeaway:** The generalized model  $O = KI + a + N(I) + K_M I_M I + K_I I_I$  captures **all** static errors. Understanding the distinct roles of  $I_I$  (shifts intercept) vs.  $I_M$  (changes slope) is essential for sensor calibration and environmental compensation.